

BSP Release Note

for Windows[®] CE 5.0

MMSP[™] 2

Release Note

MDMP2520FBSPRN-WC50

Ver. 5.4.2 / September 27,2005

MP2520F

Platform Builder for Microsoft Windows[®] CE 5.0

for MMSP 2 Application Processor

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HOW TO REACH US:

MagicEyes Digital co., Ltd.

463-824 #501, A-ONE Plaza, 269-6 Seohyeon-Dong, Bundang-Gu
Seongnam-City, Gyeonggi-Do, Korea
TEL: 82-31-788-0300
FAX:82-31-707-5765

HOME PAGE:

<http://www.mesdigital.com>

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1. Release Baseline

1.1. Software Stack

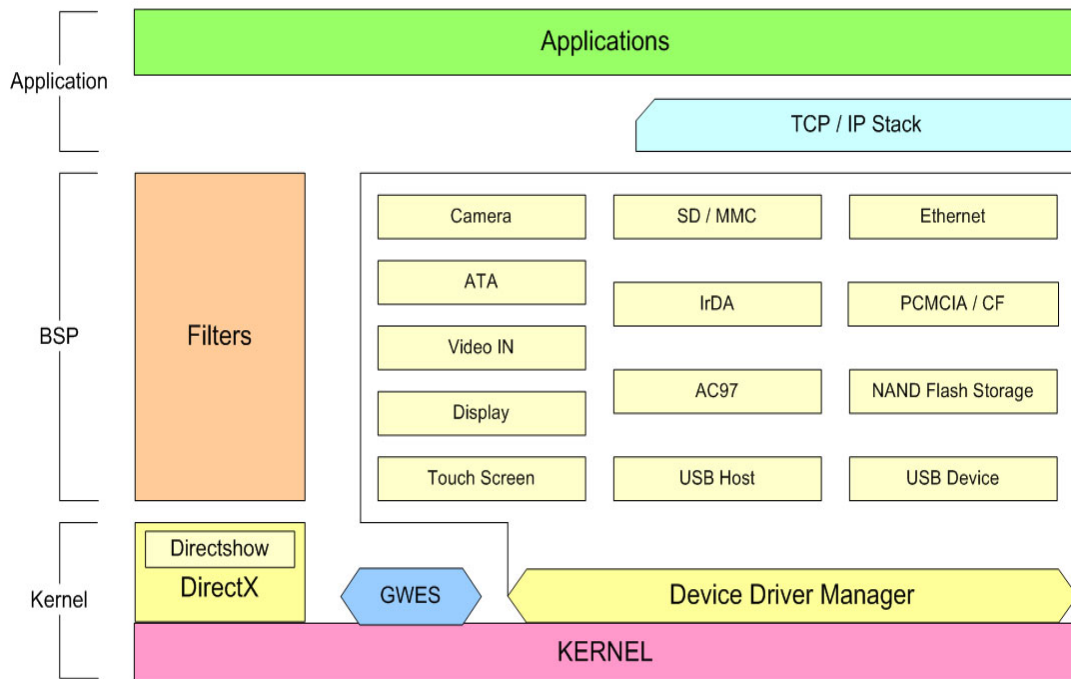


Figure 1. Software Stack

1.2. Software Modules

Basic

Kernel

The kernel is the core of the Windows CE operating system (OS). The kernel provides the base OS functionality for any Windows CE-based device. This functionality includes process, thread, and memory management. The kernel also provides some file management functionality. The kernel services enable applications to use this core functionality. MMSP2 BSP includes some kernel functions to run OS on MMSP2 based system.

Features	• OS initialization code • interrupt handling of principal peripheral • Power suspend/ wake-up • Quick rebooting (boot from valid OS image in RAM without loading process) • Remote Kernel Debugging with KITL interface
Not Supported Features	

Ethernet Bootloader

Ethernet bootloader receives Windows CE OS image (nk.bin) from platform builder through Ethernet link and load it to RAM of the MMSP2 based system. It can also store received image to the NAND flash of the system. If valid OS image is stored in NAND flash, Ethernet bootloader can load OS image from NAND flash to RAM.

Features	• Load image from Windows CE platform builder through Ethernet link • Load image from the image which is stored in NAND flash. • Quick booting mode support (To use this mode, OS image should be built with quick boot mode option.)
Not Supported Features	

Device Drivers

Display Driver

DirectDraw Display Driver

Features	● 8 / 16 / 24 bpp ● Output mode ■ LCD 640x480x60, 800x600x60, 800x480x60 ■ Conexant CX25874 ARGB : 640x480x60, 800x600x60 Composite/S-Video : 720x480x30 Component : 720x480x60 ■ Silicon Image SIL164 DVI 800x600x60 ● Hardware Overlay ■ 2 independent hardware overlay ■ YUV 4:2:2 format support ■ Fine or coarse scale up/down ■ Color control : brightness, contrast, hue, saturation ● Gamma correction ● Hardware cursor ● Bit-blit hardware acceleration ● OSD layer ● Sub-picture Layer ● Dynamic Rotation & Output Mode Change (GUI resolution is fixed.)
Sample Application	● APPS\COLORCTL ■ Sample application for color controls of overlay ● APPS\OSDDemo ■ Sample application for OSD layer ● APPS\SPUDemo ■ Sample application for SPU layer ● APPS\SetMode ■ Sample application for Dynamic Rotation & Output Mode Change. ● APPS\TestRGB ■ Sample application for controls of RGB Layer.

Omnivision OV9640 CMOS Sensor Driver

A device driver to support 1.3 mega pixel CMOS camera.

Feature	● Supports the OV9640 camera chip ● Get image data from camera and transfer to ISP ● Supports BayerRGB/CCIR656/CCIR656Like ● Supports VGA/QVGA/CIF/QCIF resolutions. ● Supports H/V mirrors. ● Supports to adjust a frame rate.
Sample Application	● FILTER\ISPSrc\MESCS ● APPS\CamView

Philips SAA7113 VideoDecoder Driver

A device driver to support a video decoder IC

Feature	● Supports the SAA7713 video decoder chip ● Get image data from external device and transfer it to ISP ● Supports S-Video/Composite inputs. ● Supports the TSCH6/PTI5NF Tuner
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Sample Application	● FILTER\ISPSource\MESVIS ● APPS\VideoIn
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Ti TVP5150A VideoDecoder Driver

A device driver to support a video decoder IC

Feature	● Supports the TVP5150A video decoder chip ● Get image data from external device and transfer it to ISP ● Supports S-Video/Composite inputs.
Sample Application	● FILTER\ISPSource\MESVIS ● APPS\VideoIn

MMSP2ISP Driver

A device driver to support ISP & SCALER module of MMSP2

Feature	● Save image data from external device to memory ● Supports interfaces to set timing parameters ● Supports YUV 4:2:2 and 2D YUV 4:2:0 format
Future release	● Provides information for AF/AE/AWB control ● Provides interfaces for color controls
Sample Application	● FILTER\ISPSource\MESCS ● FILTER\ISPSource\MESVIS ● APPS\CamView ● APPS\VideoIn

I2C Driver

A device driver to support I2C interface

Feature	● Use GPIO module to make I2C signals instead of I2C module in the MMSP2
Future release	● Supports the Hardware I2C in the MMSP2

AC97 Driver

AC97 Audio Driver to play wave audio files.

Feature	● Supports REALTEK six channel AC97 audio codec ALC650 ● Compatible with AC97 2.2
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ATA Driver

A storage device driver that manages read/write operation to ATA hard disk.

Feature	● ATA Hard Disk Driver ● Support PIO / UDMA mode Default mode is PIO. UDMA mode conflicts with with MPEG H/W dec UDMA mode can be used only for file transfer
Not Supported	● ATAPI is not supported.
Features	

Limitations	- Maximum partition size is limited to 136GB. - This ATA disk driver can recognize partitions in ATA HDD that is created by desktop Windows, but mounting partition created with this driver on desktop Windows can be failed. - If a HDD has multiple partitions, mounting some partition can be failed. In this case, only primary partition will be mounted.
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SD/MMC Storage Driver

A storage device driver that manages read/write operation to SD/MMC memory card.

Features	- SD memory card can be mounted as a storage device whenever the system runs (hot-swap of the card is supported) - Supports SD and High-speed MMC memory card.
Limitations	Some MMC card does not work.

NAND Flash Storage Driver

A storage device driver that maps part of NAND flash memory as a storage device.

Feature	Can use NAND flash memory space as a storage device except of domain that stores OS image.
Limitations	- support only 8-bit bus and size of <= 128MB - The size of NAND flash memory used for storage is fixed. - One bank with 32MB(8-bit bus, 3-addr type) NAND is tested.

USB Host Driver

A driver that manages USB devices connected to MMSP2 system. The universal serial bus (USB) is an external bus architecture for connecting a USB-compatible peripheral device such as a mouse or storage device to a host computer. USB is a communication protocol that supports serial data transfers between a host computer and a USB-compatible peripheral device.

Feature	- MMSP2 System works as an USB Host when USB Devices are connected. - Recognize mouse, keyboard, and storage device through class drivers supported by Windows CE.
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USB 1.1 Function Driver

A driver that makes MMSP2 system work as a USB device.

Feature	• USB 1.1 device controller on MMSP2 is used. • MMSP2 System works as USB device when a USB host is connected • ActiveSync link is possible with USB connection
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USB 2.0 Function Driver (Mass Storage Interface)

A driver that makes MMSP2 system work as an external mass storage when connected to a USB host.

Feature	• NET2272 USB 2.0 device controller is used. • MMSP2 System works as USB external storage device when USB host is connected
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CS8900 Ethernet Driver

A driver that facilitates communication between CS8900 Ethernet adapter card and network protocol stack provided with Windows CE operating system.

Feature	• Connect Local Area Network through Ethernet connection • Network features supported by Windows CE can be used. (Web browser, FTP server, etc.)
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PCMCIA / CF Driver

A driver that facilitates access service to PCMCIA/CF memory.

Feature	• Support PCMCIA / CF storage device. (CF/PCMCIA ATA storage driver that runs on this driver is required – Provided by Microsoft)
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Remote Controller Driver

A driver that receives input from IR remote controller and generate appropriate event.

Feature	• Sample driver for IR remote controller
Limitations	• This is not a general-purpose remote controller driver. You can use the source code of this driver for reference only.

Video Codecs

MPEG4 Decoder

MPEG4 Decoder is able to decode various types of MPEG4, such as DivX and XviD.
This CODEC supports only decoding. System layer is not included.

Feature	• DirectShow transform filter • Supported formats : DivX 3.11, DivX 4.x, DivX 5.x (SP and ASP), XviD Max. resolution : D1 (720x480) Max. bitrate : 3.5Mbps Max. frames per second : 30fps • Output format : 2D YUV
Performance	• See MPEG4 Decoder performance chart
Limitations	• Do not support 4MV + QPEL contents. • Do not support GMC which has 2 or more warping points. • In some DivX 3.11 contents you may see defections

WMV Decoder

WMV Decoder is able to decode WMV7, 8 and 9 contents
This CODEC supports only decoding. System layer is not included.

Feature	• DirectShow transform filter • Supported formats : WMV7, WMV8, WMV9 Max. resolution : QVGA (320x240) Max. bitrate : 500kbps Max. frames per second : 30fps • Output format : 2D YUV
Performance	• QVGA, 500Kbps, 30frames per second
Limitations	

Audio Codecs

Microsoft MP3 Decoder

MP3 Decoder is able to decode MP3 audio stream.

Feature	• MP3 Audio Decoder
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MES MP3 Decoder

MES MP3 Decoder evaluation version is included in BSP.

Feature	• MES MP3 Audio Decoder Eval.
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WMA Decoder

WMA Decoder is able to decode Windows Media Audio stream.

Feature	• Windows Media Audio Decoder
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Audio Capture

Audio capture filter receives audio input data for encoding.

Feature	• Audio Capture Source Filter • Capture from the microphone input • Fixed sample rate : 22050Hz
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Media File Formats

AVI

Feature	• Microsoft AVI File Format
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MPG

Feature	• MPEG1 File Format
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WMV

Feature	• WMV File Format
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ASF

Feature	• ASF File Format
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Other Media Components

DST AVI DEMUX

Feature	- Developed by DST Corporation. (Tel. +82-31-710-8800) - Demux Microsoft AVI File into audio and video stream
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DST AVI MUX

Feature	- Developed by DST Corporation. (Tel. +82-31-710-8800) - Audio and Video mux filter
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DST AVI Writer

Feature	- Developed by DST Corporation. (Tel. +82-31-710-8800) - Writes AVI multiplexed stream into a file
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Sample Applications

MESPlayer

MES Player is a simple media player for MPEG4 decoding and encoding functions.

Feature	- Play MPEG4 media file by using MPEG4 video decoder filter - Encode camera input to MPEG4 media file.
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OSImageUpdate

OSImageUpdate is a simple application program that replaces OS image in NAND flash with new image.

Feature	- Replace OS image with selected OS image file - Supports OS image file of ROM image format (nk.nb0) - Source code is in APPSIOSUpdateWin folder.
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Development Environment

TCP / IP Debugging

Remote debugging tools provided in development environment such as eMbedded Visual C++ or Windows CE Platform Builder can be connected to MMSP2 system through TCP/IP networking.

Feature	● Supports remote debugging of applications that runs on MMSP2 system through TCP/IP network (with eMbedded C++) ● Remote debugging tools are available. (ex. Remote process viewer)
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2. Installation Attention

- To avoid conflict, delete previous BSP before install current BSP.
- Import new .cec file into Platform Builder's **Manage Catalog Features** after delete previous .cec.
- Refer "**BSP Build Guide For MP2520F**" chapter of **MP2520F Development Kit User's Manual** document for information about making new platform and adding necessary components.
- For default LCD output, set DIP switch on the board as:
 - "Off Off Off Off" for DTK 4th board

3. New Components

This is the first version of Windows CE 5.0 BSP.

4. Modified Components

This is the first version of Windows CE 5.0 BSP.

5. Known Problems

5.1. USB 2.0 Mass Storage

- To use HDD as mass storage you must remove other storages from device before boot-up.

5.2. MESPlayer

- If you open contents several times MESPlayer can be unstable.

5.3. USB external storages

- There are compatibility issues. Windows CE may not recognize some thumb drives.
- We have not tested USB external HDD.

6. CODEC Status

6.1. MPEG4 Decoder

6.1.1. Supported Video Stream Formats

DivX 3.11, DivX 4.x, DivX 5.x SP, DivX 5.x ASP, XviD

- Don't support Video clips with 4mv and QPEL.
- Don't support GMC with 2 or more warping points.

6.1.2. Performance

DivX 4.x, DivX 5.x SP, XviD : 720x480, 5Mbps, 30fps

DivX 5.x ASP : 720x480, 3.5Mbps, 30fps

DivX 3.11 : 640x352, 1.8Mbps, 24fps

(for maximum performance you should use MES MP3 decoder and MES Player)

6.1.3. Known Problems

- In some DivX 3.11 contents you may see defections
- Video clips without audio may have some problems, irregular playback speed, unable to repeat, etc.
- In some video clips has very long audio chunks, you may see irregular playback after seeking.
- We have been fixed memory leakage problems in MPEG4 decoder at BSP 2.4.2 beta but it has still problem. We asked Microsoft about it. Because even we use all components provided by Microsoft it still shows memory leakage.

6.2. WMV Decoder

6.2.1. Supported WMV Versions

WMV7, WMV8, WMV9

6.2.2. Performance

QVGA (320x240), 500kbps, 30fps

6.2.3. Known Problems

none

7. Release History

This is the first release of Windows CE 5.0 BSP. For release history of Windows CE.NET 4.2 BSP, see “BSP Release Note for Windows CE.NET 4.2